



Gauge PRS Pricing

User Guide — Physical Risk Swap Valuation

MKM Research Labs

Version 1.0 — 10-April-2026

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1 Introduction

A **Physical Risk Swap** (PRS) is a credit-derivative-style contract that transfers flood risk between two parties. The protection buyer (payer) pays a periodic premium in exchange for a contingent payment if a specified flood event occurs at a river gauge. The structure mirrors a single-name Credit Default Swap (CDS), with the “credit event” replaced by a physical flood event.

1.1 Why PRS Exists

Property owners, mortgage lenders, and insurers face concentrated exposure to river flooding. A PRS allows them to hedge this exposure by transferring the risk to a willing counterparty (hedge fund, reinsurer, or capital markets investor) in exchange for a spread. The spread compensates the protection seller for the probability of having to make a payout.

1.2 Gauge vs. Property PRS

A **Gauge PRS** references a single river gauge. The trigger is defined by a water level threshold (alert, warning, or severe) at that gauge. The pricing uses an analytical hazard-rate model.

A **Property PRS** references a specific property and its nearest gauges. It is priced from Monte Carlo event counts rather than analytical hazard rates. This guide covers gauge PRS; property PRS is documented separately.

2 From Hazard Rate to Fair Spread

The pricing pipeline converts an annual flood probability into a fair spread in basis points (bps). This section walks through each step.

2.1 Annual Probability to Continuous Hazard Rate

The gauge hazard curve provides an **annual hazard rate** — the probability that a flood event occurs within one year. For example, a 5% annual probability at the severe trigger level.

The CDS pricing framework requires a continuous hazard rate λ , converted via:

$$\lambda = -\ln(1 - p_{\text{annual}})$$

For $p = 0.05$: $\lambda \approx 0.0513$.

2.2 Survival Probability

The survival probability $S(t)$ at time t (years) is the probability that no flood event has occurred by time t :

$$S(t) = e^{-\lambda t}$$

For $\lambda = 0.0513$ and $t = 5$: $S(5) = e^{-0.2565} \approx 0.7738$.

2.3 Risky Annuity

The risky annuity is the present value of £1 paid quarterly, conditional on no flood event having occurred, discounted at the risk-free rate:

$$A = \sum_{i=1}^n \Delta t \cdot S(t_i) \cdot D(t_i)$$

where $\Delta t = 0.25$ (quarterly), $n = 4 \times \text{tenor}$, and $D(t) = e^{-r(t) \cdot t}$ is the discount factor from the yield curve.

A typical 5-year risky annuity is in the range 3.5–5.0, depending on the hazard rate and yield curve. The risky annuity is the denominator of the CDS spread formula and also serves as the DV01 (see Chapter 4).

2.4 Protection Leg

The protection leg is the present value of the payout made upon a flood event, weighted by the probability of default in each period:

$$P = \sum_{i=1}^n (1 - R) \cdot (S(t_{i-1}) - S(t_i)) \cdot D(t_{\text{mid},i})$$

where R is the recovery rate (default 0.0 for flood events, meaning full loss) and $t_{\text{mid},i}$ is the midpoint of period i .

2.5 Fair Spread

The fair spread equates the premium leg with the protection leg:

$$s_{\text{fair}} = \frac{P}{A}$$

Expressed in basis points:

$$s_{\text{bps}} = s_{\text{fair}} \times 10,000$$

2.6 Worked Example

Input	Value
Annual hazard rate (severe)	5%
Tenor	5 years
Recovery	0%
Risk-free rate	3% flat

1. $\lambda = -\ln(0.95) = 0.0513$
2. Risky annuity $A \approx 4.71$
3. Protection leg $P \approx 0.0245$

4. Fair spread = $0.0245/4.71 = 0.00520 = 52$ bps

The protection buyer would pay 52 basis points per annum on the notional to hedge severe flood risk at this gauge.

3 Trade Structure

3.1 Payer vs. Receiver

Payer (protection buyer) Pays the periodic spread. Profits when flood risk increases (spread widens). Long risk.

Receiver (protection seller) Receives the spread. Profits when flood risk decreases (spread tightens). Short risk.

3.2 Key Terms

Field	Typical Value	Description
Notional	£10M	Face value; maximum payout
Tenor	1, 2, 3, or 5Y	Contract length
Trigger Level	severe	Water level threshold (alert/warning/severe)
Trade Spread	45 bps	Premium agreed at inception
Day Count	ACT/360	ISDA convention
Payment Freq	6M	Semi-annual premium payments
Recovery	0%	Full loss on flood event
Currency	GBP	Denomination

3.3 Trigger Levels

Each gauge has three statutory flood levels (metres above ordnance datum):

Flood Alert Early warning — be prepared. Lowest threshold.

Flood Warning Flooding expected — take action.

Severe Flood Warning Severe flooding — danger to life. Highest threshold.

The trigger level determines which hazard rate drives the spread. Severe triggers have lower probabilities (rarer events) and therefore lower spreads.

3.4 Yield Curve

The risk-free yield curve used for discounting is a term structure of continuously compounded rates. The default curve is:

Tenor	Rate
1Y	3.5%
2Y	4.0%
3Y	4.3%
4Y	4.5%
5Y	4.0%

Rates between tenors are linearly interpolated. Adjusting the yield curve changes discount factors and therefore the risky annuity and fair spread. A parallel 50 bp shift in rates typically moves the fair spread by 1–3 bps.

4 Greeks & Sensitivity

4.1 FS01 — Flood Sensitivity

FS01 is the P&L change from a 1 basis point increase in the annual hazard rate. It is computed by bumping:

$$\text{FS01} = \frac{\Delta s}{10,000} \times A \times N$$

where $\Delta s = s(p + 0.01\%) - s(p)$ is the spread change in bps from a 1 bp hazard rate bump, A is the risky annuity, and N is the notional.

Example: If $\Delta s = 0.75$ bps and $A = 4.71$ on £10M:

$$\text{FS01} = \frac{0.75}{10,000} \times 4.71 \times 10,000,000 = \text{£}3,532$$

Interpretation: if the gauge's annual severe flood probability increases by 1 bp (e.g. 5.00% to 5.01%), a payer's position gains £3,532 and a receiver's position loses the same.

4.2 Risky Annuity as DV01

The risky annuity A also serves as the spread DV01. A 1 bp change in the *traded spread* (not hazard rate) produces a P&L of:

$$\text{DV01} = \frac{1}{10,000} \times A \times N$$

For $A = 4.71$ on £10M: $\text{DV01} = \text{£}4,710$ per bp.

4.3 Direction Convention

	Spread widens	Spread tightens
Payer (buyer)	Gains	Loses
Receiver (seller)	Loses	Gains

5 Mark-to-Market

The mark-to-market (MTM) of an open trade is:

$$\text{MTM} = \frac{s_{\text{fair}} - s_{\text{trade}}}{10,000} \times A \times N \times d$$

where $d = +1$ for a payer and $d = -1$ for a receiver.

5.1 Worked Example: Day 1 to Day 2

	Day 1	Day 2
Annual hazard rate	5.00%	4.50%
Fair spread	52 bps	48 bps
Trade spread	45 bps	45 bps
Risky annuity	4.71	4.71
MTM (payer, £10M)	£32,970	£14,130
Daily P&L	—	−£18,840

The payer's position lost £18,840 because the hazard rate tightened (flood risk decreased), narrowing the fair spread towards the trade spread.

6 Parameter Reference

Parameter	Default	Increase	Decrease
TENORS	[1,2,3,5]	More maturity buckets	Fewer contract options
NOTIONALS	£5–20M	Larger face value, larger FS01	Smaller face value
RECOVERY	0.0	Reduces protection leg PV	Higher LGD, higher spreads
DEFAULT_YIELD_CURVE	3.5–4.5%	Lower annuity, higher spread	Higher annuity, lower spread
BUMP_1BP	0.0001	Larger bump for FS01	Finer sensitivity
MIN_PRS_SPREAD_BPS	2.0	Higher floor on fair spread	Lower floor (near-zero risk)
TRIGGERS	[severe]	More trigger levels available	Fewer trigger options
SPREAD_OFFSET_MIN	3.0 bps	Wider bid-ask spread	Tighter execution

Parameter	Default	Increase		Decrease	
SPREAD_OFFSET_MAX	10.0 bps	More execution slip-		Tighter max offset	
		page			
EA_FLOOD_ZONE_RATES	0.1–5%	Higher	zone-based	Lower	zone-based
		hazard		hazard	

Document History

Version	Date	Author	Description
1.0	10-Apr-2026	MKM Research Labs	Initial release